Multilevel Features Fused and Change Information Enhanced Neural Network for Hyperspectral Image Change Detection

Jigang Ding, Xiaorun Li, Shu Xiang, and Shuhan Chen, *Member, IEEE*

***Abstract*— Hyperspectral image change detection (HSI CD) refers to identifying and analyzing differences between two HSIs acquired in the same area but at different times. However, current deep learning (DL)-based methods have limitations in fully exploiting the change information between bitemporal images and utilizing multilevel features. To address these issues, we propose a novel Multi-level features Fused and Change information Enhanced neural Network (MFCEN) for HSI CD. The proposed MFCEN method leverages the hierarchical low- and high-level features of bitemporal images, allowing for the direct capture and enhancement of change features. First, a Siamese-based network is employed to extract multilevel features from the bitemporal images, including low-level spatial details and high-level semantic features. Within the temporal change information branch (TCIB) at each level, the change features are reinforced by the semantic features of each image, and the change features act as a guiding force to direct the feature extraction of each bitemporal image to focus more on the change region. Next, the enhanced change features of each level are fed into the multilevel features fusion module (MFFM) to aggregate the fine-grained details and high-level semantics. Finally, the fused features, enriched with multilevel change information, are utilized for CD. The experiments demonstrate that our proposed MFCEN outperforms existing methods on three public datasets. The code will be available at https://github.com/Ding201901/MFCEN.**

***Index Terms*— Attention mechanism, change detection (CD), hyperspectral image (HSI), multilevel features.**

I. Introduction

As THE economy and society continue to evolve, the need for precise and efficient monitoring of the Earth's surface in response to human activities, natural disasters, and environmental changes has become growing urgent. Change detection (CD) plays a vital role in monitoring land use and land cover. CD provides detailed change information on the Earth's surface by analyzing images obtained at different times within the same geographic region. With the advancement of imaging technology, remote sensing images have become a valuable source of ground object information, particularly hyperspectral images (HSIs). HSIs offer numerous bands and almost continuous spectral information, enabling more accurate ground object analysis. As a result, HSI CD has gained increasing attention in recent years and has found extensive application in diverse fields, including crop monitoring [1], natural disaster analysis [2], and land cover assessment [3].

During the early years, traditional methods were primarily proposed to monitor the land cover changes. Image algebra-based methods detect changes in surface objects by performing mathematical operations on the differences between pixels, such as subtraction and division. Change vector analysis (CVA) [4] computes the change vector between hyperspectral data at two different times to measure the difference. Later, techniques like Sequential Spectral CVA [5] and Robust CVA [6] were proposed to improve the performance of CVA. Image transformation methods aim to mitigate the impact of unchanged regions by transforming the hyperspectral data into another feature space. Iteratively Reweighted Multivariate Alteration Detection (IR-MAD) [7] focuses in successive iterations on observations with uncertain change status by weighting based on a no-change measure. The classification-based method obtains change maps by comparing the classification results of each image. While this method can provide more detailed information about changes, the accuracy of the detection performance is highly reliant on the quality of the classification results. In addition, other advanced methods, such as conditional random fields [8], tensor decomposition and reconstruction detector (TDRD) [9], and multiple morphological profiles [10], have been successfully applied to CD. These traditional methods exact handcrafted features without any prior information, limiting the further improvement of detection results.

Recently, deep learning (DL) has demonstrated remarkable performance in computer vision (CV) and remote sensing with its excellent ability to extract features. Therefore, lots of DL-based methods for CD are proposed. Generally, the DL-based methods can be divided into early-fusion (EF) and late-fusion (LF) methods due to the difference in temporal information utilization, as presented in Fig. 1.

The EF method obtains the fused image by making subtraction or concatenation and feeds it into a single-branch network. In [11], to focus more on change intensity, the difference image undergone an absolute value transformation before the extraction of features. Ou et al. [12] constructed a difference matrix as input to mine the inner relationship between different

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The authors are with the College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: 12010075@zju.edu.cn; lxyly@zju.edu.cn; 12210076@zju.edu.cn; 11410057@zju.edu.cn).

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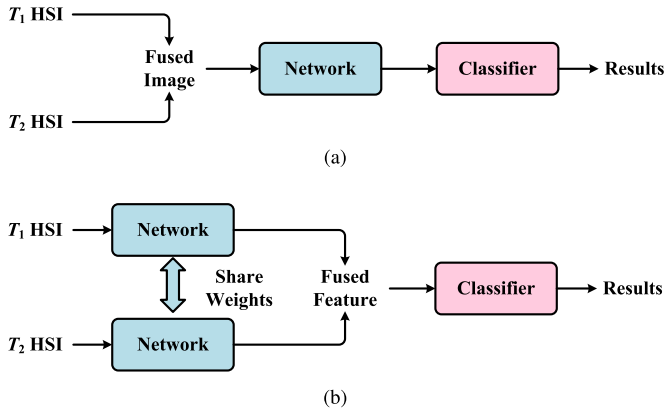


Fig. 1. Illustration of (a) EF method and (b) LF method.

bands of pixel pairs. In [13], the concatenated image was fed into a residual self-calibrated network to capture the discriminative features at different scales. Song et al. [14] presented a CD framework integrating iterative active learning and affinity graph learning, where the difference image was used as input. These EF methods input the concatenation or subtraction of bitemporal images into a single-branch network, leading to difficulties in extracting deep semantic features of the respective images.

The LF methods have gained more attention in recent years. They take each bitemporal image as input and improve the accuracy of CD by extracting the high-level features of each image. A Siamese network is a neural network architecture containing two or more subnetworks with the same weights and architecture. By comparing the embeddings of two inputs, Siamese networks can identify differences between the two inputs that may indicate a change. Based on the autoencoder, Zhao et al. [15] concatenated the encoder-extracted features for each time to be utilized for CD. To reduce noise and retain spatial information, Zhan et al. [16] utilized Euclidean distance to measure the similarity of the tensor pairs without needing category information. In [17], the change information is captured by concatenating the features of each image, which are extracted using CNNs in multiple forms. Song et al. [18] introduced novel bidirectional spectral coding and enhanced spatial residual designs to provide useful feature sources. Nevertheless, the mere application of subtraction and concatenation to bitemporal features makes it challenging to fully exploit the change information. Recurrent neural networks (RNNs) are introduced to CD since they can effectively process sequence data. In both the recurrent 3-D fully convolutional network (Re3FCN) [19] and recurrent CNN (ReCNN) [20], the CNN-extracted representations are input to long short-term memory (LSTM) to learn change patterns between the multitemporal images. Although LF methods can capture deep features of each image and enhance the utilization of change information through RNNs, they exhibit a deficiency in attending to change regions during feature extraction, leading to failure to fully exploit the temporal dependency.

The attentional mechanism can help the neural network to concentrate on key information, enhancing model performance and efficiency. Some methods have introduced attention mechanisms to enhance the precision of HSI CD. Gong et al. [21] used several fully connected layers to compute a 2-D Gaussian distribution, from which the attention weight of the spatial location was sampled. Huang et al. [22] proposed a parallel spectral-spatial attention (SA) module to make the features more discriminative while preserving the dependence of the original data. Song et al. [23] proposed a cross-temporal interaction symmetric attention network (CSANet), which employed self-attention and cross-temporal attention to enhance the various features. Note that our proposed method is based on a three-branch structure and utilizes multilevel features, which differs from [23]. Li et al. [33] applied SA to image patches of varying sizes, ensuring that the central pixel of the reconstructed patches receives the highest weight while the surrounding pixels contribute only the essential spatial information. Convolutional block attention module (CBAM) [24] and its variants have been introduced to HSI CD to exact more comprehensive information about the change region [25], [26], [27].

Transformer models [28], [29], equipped with self-attention, have been widely adopted in natural language processing and CV tasks because of their ability to capture long-range dependencies in a sequence. Wang et al. [30] proposed a spectral-spatial-temporal transformer (SST-Former), which used a Siamese network composed of transformers to model spectra and space dependency of each image sequentially and utilized a transformer-based subnetwork with the residual structure to capture the change information. Note that our proposed method is structure differs from [30]. Our proposed MFCEN constructs a three-branch network to extract semantic features of bitemporal images and change features simultaneously. In [31], the pixel sequences of each bitemporal image were concatenated as input to the transformer encoders, leveraging the long-range spatial-temporal dependencies. Liu et al. [32] and Li et al. [33] used the 2-D self-attention mechanism to improve the feature representation and select effective information. Later, the cross-attention mechanism was introduced to enhance the utilization of temporal change information. In [34], multiple layers of hybrid multihead attention (MHA) were utilized to map multitemporal image differences to abundance matrices and hierarchically extract enhanced difference features. Based on MHA, Wang et al. [35] proposed a triplet transformer architecture, which employed parents-temporal attention model temporal dependency and brother-SA to capture spatial correlations.

Despite the attention mechanism's ability to enhance significant information and suppress irrelevant ones, it is notable that most of these methods still belong to either EF or LF methods. While some methods incorporate cross-attention mechanisms to capture the temporal dependency between bitemporal images, they can only realize the interaction of the change information rather than extracting and enhancing it directly. Therefore, there is still a need for further investigation on how to effectively leverage the deep features and temporal dependency of bitemporal images.

In recent years, various approaches have been explored to exploit features at multiple levels to optimize feature utilization and enhance the accuracy of CD. Hu et al. [36] employed

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efficient multihead self-attention (MHSA) to eliminate the redundant information from various levels of feature representations and then concatenated these refined features for CD. In [37], the multipath convolutional LSTM (ConvLSTM) block is used to learn multilevel temporal dependencies between HSIs to capture changes at different spectral scales. Yang et al. [38] utilized ConvLSTM to integrate temporal change information within multiscale features in each level, where the multiscale features were extracted by convolution kernels of different sizes. Qu et al. [39] presented a multilevel encoder–decoder attention network (ML-EDAN), where LSTM is used to learn the temporal dependency at each level, and then the multilevel change features were concatenated for CD. Also based on the subnetwork of encoder–decoder, [27] learned the subtle difference and enhanced the critical change information from different scales. While these methods offer novel insights for addressing the HSI CD task, they only utilize different levels of features by concatenation, which tends to result in unbalanced feature utilization and potential loss of crucial information. Therefore, there is room for improvement in taking full advantage of the correlations and differences between multilevel features.

Although DL-based methods have made significant achievements in HSI CD, there are still certain limitations that can be summarized as follows.

1) Underutilization of temporal dependency. The EF methods overlook the high-level semantics of HSIs, and the LF methods neglect the utilization of change information during feature extraction. Therefore, these methods have difficulty in completely constructing the nonlinear change rule between bitemporal images. While some methods incorporate cross-attention to enhance the utilization of temporal dependency, they primarily facilitate the interaction of change information rather than directly capturing and enhancing change features. Therefore, effectively leveraging temporal dependency to improve the network's ability to capture change information remains a challenging task.
2) Insufficient utilization of multilevel information. Most existing DL-based approaches can only utilize the features of one layer, either the input or output of the network, leading to the loss of valuable information. Although some methods attempt to incorporate different levels of features through concatenation and skip connections, this often results in unbalanced feature utilization and potential loss of crucial information. Consequently, it remains challenging to fully explore the correlation and differences between different levels of features to effectively integrate detailed spatial features and abstract semantic information.

To address these limitations, we propose a novel Multilevel features Fused and Change information Enhanced neural Network (MFCEN) for HSI CD. A Siamese-based network is utilized to capture the multilevel features of bitemporal images, including detailed spatial features and abstract semantic information. In each level of the temporal change information branch (TCIB), the change features are strengthened by the semantic features of each bitemporal image, and the change features serve as a guiding force to direct the features of each image to focus more on the change region. The enhanced multilevel change features are fed into the multilevel features fusion module (MFFM) to aggregate the multigranularity representations. Finally, the fused features enriched with multilevel change information are used for CD. In this article, the main contributions are highlighted as follows.

1) We propose a novel MFCEN for HSI CD, which enables the leverage of the hierarchical low- and high-level features from bitemporal HSIs and directly captures the change information level by level, improving the CD performance of the framework.
2) The TCIB is designed to directly extract and enhance change features from the difference image at various levels. The band-wise image fusion strategy (BIFS) is first introduced to obtain the difference image used to directly extract the change features. BIFS leverages spatial differences between corresponding bands by group convolution. Then, via the temporal attention enhanced module (TAEM) embedded in TCIB, the change information of change features is strengthened by the inherent characteristics of each bitemporal image, and the features of each image are guided by the change features to focus more on the change regions.
3) The transformer is employed for multilevel feature fusion in HSI CD for the first time. The proposed MFFM based on transformer encoders effectively aggregates multilevel change features, such as spatial details and abstract semantic information. By learning the importance and intrinsic relevance of different features, a more discriminative change representation is obtained.

The organization of the remaining sections is as follows. Section II comprehensively explains our proposed MFCEN framework. Section III validates the performance of our proposed method through experiments on three datasets. Finally, Section IV draws the conclusion.

II. Proposed Method

This section elaborates on the proposed MFCEN framework for HSI CD. As illustrated in Fig. 2, MFCEN consists of a Siamese architecture to extract hierarchical features, a temporal change information branch (TCIB) to extract and enhance change features level by level, and a feature fusion module to aggregate multilevel enhanced change features.

A. Siamese Network

The Siamese-based network is designed to capture the multilevel spatial-spectral features. Embedded with the CBAM-BP, the residual connection block (ResBlock) in different layers extract multilevel features, including precise spatial details and abstract semantic information. The ResBlock, comprising the basic block of the original ResNet [40] with the addition of our CBAM-BP, is introduced to extract spatial-spectral features of each time phase. Fig. 3 shows the detailed structure of ResBlock. Given the HSI patch $X_t \in \mathbb{R}^{B\times H\times W}$ $(t = 1, 2)$ as the input, a Convolutional Block (ConvBlock) is first employed to extract the primary features. This block consists of a 2-D

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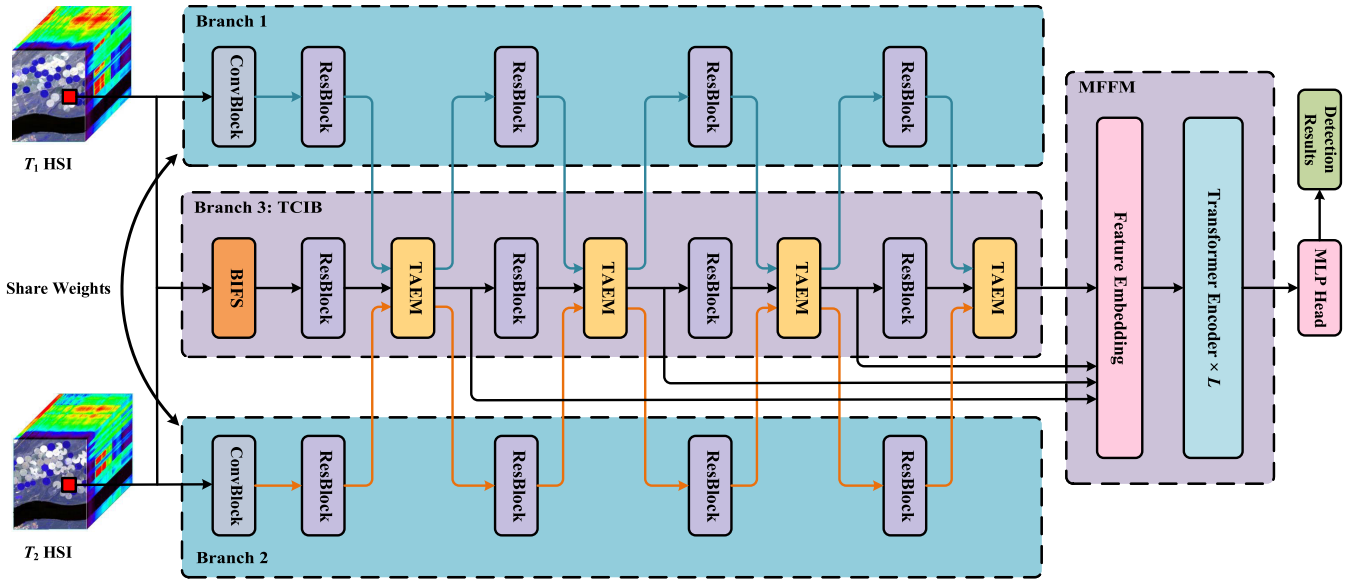


Fig. 2. Verview of our proposed method. First, a Siamese-based network is employed to extract the multilevel features of the bitemporal images. Second, in each level of the TCIB, the designed TCIB is employed to strengthen the change information and surpasses the relevant ones. Finally, the multilevel features are aggregated in MFFM based on transformer encoders.

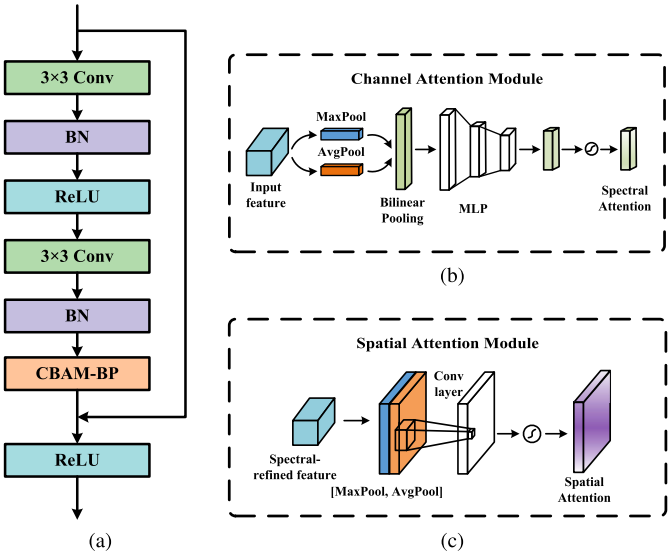


Fig. 3. Structure of (a) ResBlock, (b) CA, and (c) SA.

convolutional layer with the same spatial size as the patch, a batch normalization (BN) layer, and a rectified linear unit (ReLU) activation function

$$F_t = \mathrm{ReLU}\left(\mathrm{BN}\left(f^{H\times W}(X_t)\right)\right) \tag{1}$$

where $F_t \in \mathbb{R}^{C\times H\times W}$ denotes the primary feature of each bitemporal image, C is the channel dimension of the feature maps, and $f^{H\times W}(\cdot)$ denotes the 2-D convolution with a kernel size of $H \times W$. Within the ResBlock, the features F_t are initially utilized to extract coarse features, resulting in the output F_t'

$$F_t' = \mathrm{BN}\left(f^{3\times 3}\left(\mathrm{ReLU}\left(\mathrm{BN}\left(f^{3\times 3}(F_t)\right)\right)\right)\right). \tag{2}$$

To enhance the model's ability to prioritize important feature information, an improved version of the convolutional block attention module is integrated into the block. CBAM-BP combines channel attention (CA) and SA mechanisms to perform adaptive weighting of the input feature maps in both the channel and spatial domains, as illustrated in Fig. 3(b) and (c), respectively.

Considering hyperspectral data is characterized by numerous bands and rich spectral information, we propose an enhancement to the original CA mechanism using bilinear pooling. Specifically, the feature maps are first transformed into $1 \times 1 \times C$ feature vectors using global max and average pooling, respectively, after which the bilinear pooling is used to aggregate the representations of the two feature vectors. Then, a multilayer perceptron (MLP) with one hidden layer learns interchannel relationships to generate channel weights. Finally, the sigmoid function σ is applied to normalize the CA vector within a range of 0–1. The CA can be summarized as follows:

$$M_{t,\mathrm{ca}} = \sigma\left(\mathrm{MLP}\left(\mathrm{BP}\left(\mathrm{AvgPool}\left(F_t'\right), \mathrm{MaxPool}\left(F_t'\right)\right)\right)\right) \tag{3}$$

where $M_{t,\mathrm{ca}}$ is the CA vector, and BP denotes bilinear pooling with the computational procedure as follows:

$$\xi\left(F_t'\right) = \mathrm{vec}\left(\mathrm{AvgPool}\left(F_t'\right)^T \mathrm{MaxPool}\left(F_t'\right)\right) \tag{4}$$

$$u_t = \mathrm{sign}\left(\xi\left(F_t'\right)\right)\sqrt{\left|\xi\left(F_t'\right)\right|} \tag{5}$$

$$z_t = \frac{u_t}{\|u_t\|_2} \tag{6}$$

where $\xi(F_t') \in \mathbb{R}^{C^2\times 1}$ is the bilinear feature vector, and vec denotes converting a matrix into a vector. The vector $\xi(F_t')$ is processed by a signed square root operation and l_2 normalization. The final output z_t of BP is used for the following attention computation. Incorporating BP enables the improved CA to jointly leverage both pooling techniques for superior spectral mining. Specifically, average pooling extracts overall semantics, while max pooling highlights significant local features. BP combines both holistically, learning integrated global–local representations.

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Subsequently, the 1-D CA map $M_{t,\mathrm{ca}}$ is element-wise multiplied with the input feature F_t' to generate the refined spectral feature $F_{t,\mathrm{ca}}$, where important channels will be given greater weight. The computation process can be described as follows:

$$F_{t,\mathrm{ca}} = M_{t,\mathrm{ca}} \otimes F_t' \tag{7}$$

where $\otimes$ denotes element-wise multiplication.

Fig. 3(c) provides a detailed illustration of SA. First, for the input feature map, max pooling and average pooling operations are performed in channel dimensions to obtain descriptors for structural and global features of the space, respectively. Next, a 2-D 7×7 convolutional layer is employed to integrate these two types of information, followed by the sigmoid function to transform the output into a SA map $M_{t,\mathrm{sa}}$. Finally, the SA map is multiplied element-wise with the input feature map to generate enhanced spatial features $F_{t,\mathrm{ca}}$, where the informative region is enhanced and useless ones are suppressed. Formally,

$$M_{t,\mathrm{sa}} = \sigma\left(f^{7\times7}\left(\left[\mathrm{AvgPool}\left(F_{t,\mathrm{ca}}\right); \mathrm{MaxPool}\left(F_{t,\mathrm{ca}}\right)\right]\right)\right) \tag{8}$$

$$F_{t,\mathrm{sa}} = M_{t,\mathrm{sa}} \otimes F_{t,\mathrm{ca}}. \tag{9}$$

Next, the attention-enhanced features $F_{t,\mathrm{sa}}$ are combined with the original input F_t through residual connections to alleviate the issues of gradient vanishing and gradient exploding. The ultimate output can be formulated as

$$\begin{cases} F_t^i = \mathrm{ReLU}\left(F_{t,\mathrm{sa}}^1 + F_t\right), & i = 1 \\ F_t^i = \mathrm{ReLU}\left(F_{t,\mathrm{sa}}^i + \mathrm{EF}_t^{i-1}\right), & i = 2, 3, 4 \end{cases} \tag{10}$$

where F_t^i denotes the feature obtained by the ith ResBlock of Branch t ($t = 1, 2$). EF_t^{i-1} ($i = 2, 3, 4$) is the input of ith ResBlock, obtained by the temporal attention enhancement module of the layer $i-1$. The calculation process of EF_t will be introduced thoroughly in Section II-B2.

B. Temporal Change Information Branch

To extract and enhance change features directly and facilitate the interaction of change information, we introduce Branch 3, called the TCIB, as shown in Fig. 2. The overall structure of TCIB is similar to the Siamese network mentioned above, except for two components: BIFS and TAEM. In the subsequent sections, we will elaborate on these components.

1) Band-Wise Image Fusion Strategy: Due to the varying imaging conditions of different remote sensing images, such as lighting and viewing angle, the differences between bitemporal images cannot be effectively characterized by simple mathematical operations, such as subtraction and concatenation. Moreover, the incorporation of spatial information can effectively mitigate the influence of noise. Building upon this, we introduce the BIFS to obtain fused image enriched with more discriminative change information, where group convolution is employed to leverage the spatial information between corresponding bands. The diagram of BIFS is shown in Fig. 4.

Given the HSI patches $X_t = (x_{t,1}, \ldots, x_{t,B}) \in \mathbb{R}^{B\times H\times W}$, we concatenate the corresponding bands of the bitemporal image band-by-band. Next, we employ 2-D group convolution to

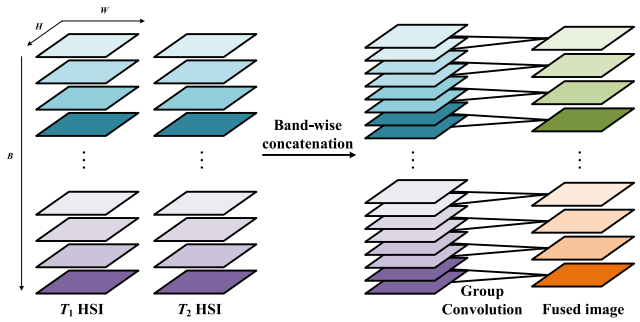


Fig. 4. Diagram of the BIFS.

fuse the corresponding bands and generate the fused difference image. The calculation process can be described as follows:

$$F_3^0 = \mathrm{GELU}\left(\mathrm{BN}\left(f_{\mathrm{group}}^{3\times3,B}\left(x_{1,1}, x_{2,1}, \ldots, x_{1,B}, x_{2,B}\right)\right)\right) \tag{11}$$

where $F_3^0 \in \mathbb{R}^{C\times H\times W}$ is the fused difference image, $f_{\mathrm{group}}^{3\times3,B}(\cdot)$ denotes the 2-D group convolution with a kernel size of 3×3 and the group size of B, and GELU is the Gaussian Error Linear Unit.

Afterward, F_3^0 is fed into the subnetwork consisting of ResBlocks to directly extract change features; the subnetwork is not parameter-shared with the previously mentioned Siamese-based network. Additionally, each ResBlock in the TCIB is followed by an embedded temporal attention enhanced module (TAEM) to interact and enhance change information.

2) Temporal Attention Enhanced Module: The fused difference image obtained using BIFS incorporates original information from each bitemporal image. However, the difference image lacks high-level semantic features from each bitemporal image, which could enhance the change information. Moreover, the Siamese network relies on the change information to effectively guide the model's focus toward the change region. To address these limitations, we design the TAEM based on the multihead cross-attention mechanism to facilitate the interaction of change information at various levels, as illustrated in Fig. 5.

Given the feature maps F_1^i, F_2^i, and F_3^i of different branches, a 2-D convolutional layer is first employed to generate the queries (Q), keys (K), and values (V)

$$Q_t^i, K_t^{i,h}, V_t^{i,h} = F_t^i W_t^{Q,i,h}, F_t^i W_t^{K,i,h}, F_t^i W_t^{V,i,h} \tag{12}$$

where $Q_t^{i,h}$, $K_t^{i,h}$, and $V_t^{i,h}$ denote the hth queries, keys, and values of F_t^i. $W_t^{Q,i,h}$, $W_t^{K,i,h}$, and $W_t^{V,i,h}$ are the weights of the 2-D convolution. Next, temporal attention is performed to enhance the features' change information of each branch. The TAEM process can be defined as follows:

$$\begin{cases} F_1^{i,h} = \mathrm{softmax}\left(\dfrac{Q_1^{i,h} K_3^{i,h^T}}{\sqrt{d_k}}\right) V_1^{i,h} \\ F_2^{i,h} = \mathrm{softmax}\left(\dfrac{Q_2^{i,h} K_3^{i,h^T}}{\sqrt{d_k}}\right) V_2^{i,h} \\ F_3^{i,h} = \mathrm{softmax}\left(\dfrac{Q_3^{i,h} K_1^{i,h^T} - Q_3^h K_2^{h^T}}{\sqrt{d_k}}\right) V_3^{i,h} \end{cases} \tag{13}$$

where $F_t^{i,h}$ is the hth ($h = 1, 2, \ldots, H$) attention map of ith ResBlock in branch t. Subsequently, the attention heads are

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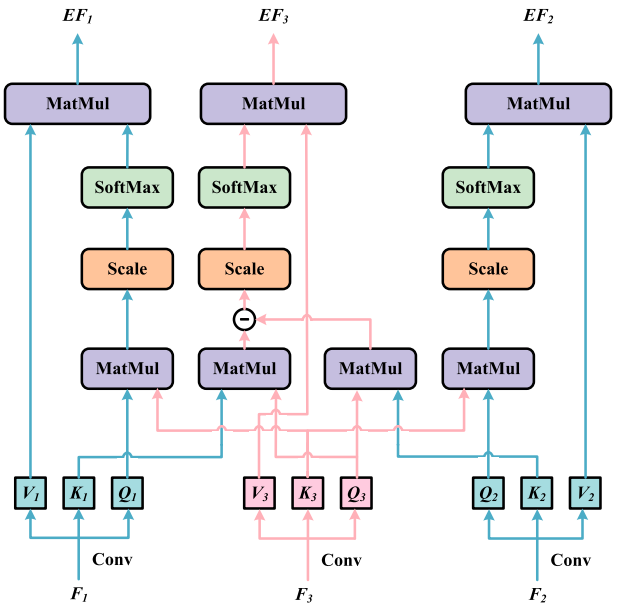


Fig. 5. Overview of the TAEM.

concatenated and fed to a 1×1 2-D convolution to generate the enhanced features EF_t^i of each branch

$$\mathrm{EF}_t^i = f^{1\times 1}\left(\mathrm{Concat}|_{h=1}^{H}\left(F_t^{i,h}\right)\right). \tag{14}$$

Notably, the enhanced feature EF_t^i obtained at layer i is the ResBlock's input of layer $i+1$, as illustrated in (10).

The proposed TAEM facilitates the enhancement of change information within the change features by leveraging the semantic features of the bitemporal images. Simultaneously, the features from each temporal phase concentrate more on the change region with the assistance of the change features.

C. Multilevel Features Fusion Module

The enhanced change features $\mathrm{EF}_3^i (i = 1, 2, 3, 4)$, generated from various layers of the TAEM module, comprise both low-level coarse spatial details and high-level semantic information. To obtain the final discriminative change features, we design a multilevel features fusion module based on transformer encoders to learn and aggregate the multigranularity features.

1) Feature Embedding: Group convolution is utilized to convert the features at different levels into feature embeddings. The convolution kernel size is equal to the feature map size, and no padding is necessary. Formally,

$$E = \left[\mathrm{CLS}; \mathrm{Concat}|_{i=1}^{4}\left(f_{\mathrm{group}}^{H\times W,d}\left(\mathrm{EF}_3^i\right)\right)\right] \tag{15}$$

where $E \in \mathbb{R}^{5\times d}$ denotes the feature embedding sequence with the dimension of d, and $\mathrm{CLS} \in \mathbb{R}^{1\times d}$ is the class token, which is randomly initialized and appended at the start of the sequence to gather change information from various levels of features. $f_{\mathrm{group}}^{H\times W,d}(\cdot)$ denotes the 2-D group convolution with a kernel size of 3×3 and the group size of d. In this way, we convert the feature maps with different dimensions across multiple layers into vectors with the same dimension, effectively leveraging the spatial information within each channel.

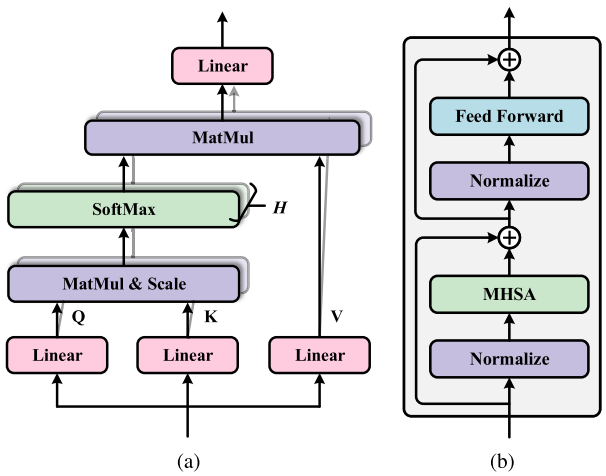


Fig. 6. Illustration of (a) MHSA and (b) transformer encoder.

2) Multilevel Features Fusion: Due to its unique architecture, the transformer has emerged as a powerful tool for handling sequence tasks. Transformer employs a self-attention mechanism that allows it to attend to all positions in the sequence in spite of their relative position. This enables the model to capture complex relationships between distant elements in the sequence. Therefore, the transformer encoders are employed to integrate multilevel features and generate more discriminative change features. As demonstrated in Fig 6, each transformer encoder comprises a multihead self-attention (MHSA) mechanism, an FFN layer, layer normalization (LN), and residual connection. Specifically, the output of the lth ($l = 1, 2, \ldots, L$) transformer layer can be mathematically represented when the feature embedding E is provided as input of transformer encoders

$$X_l' = \mathrm{MHSA}(\mathrm{LN}(X_{l-1})) + X_{l-1} \tag{16}$$

$$X_l = \mathrm{FFN}\left(\mathrm{LN}\left(X_l'\right)\right) + X_l' \tag{17}$$

where X_l denotes the output of lth transformer layer and MHSA can be formulated as

$$\mathrm{MHSA}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{18}$$

where Q, K, and V denote the matrices of queries, keys, and values, respectively. d_k is the dimension of the keys. Q, K, and V can be calculated by the following process:

$$Q, K, V = XW^Q, XW^K, XW^V \tag{19}$$

where W^Q, W^K, and W^V are the learned parameters.

Our proposed MFFM leverages multilevel change information from the CNN to achieve an effective fusion of multigranularity features through transformer's MHA mechanism. Specifically, the low-level features contain more local texture information, while the high level contains more global abstract semantics. The MHSA in transformer learns the dependencies between the hierarchical features and learns the representation of each feature subspace to focus on different feature combinations. Compared with simple concatenation, the self-attention mechanism learns the weights of different features for more effective feature fusion. Therefore, the fused features not only contain rich semantic information but also

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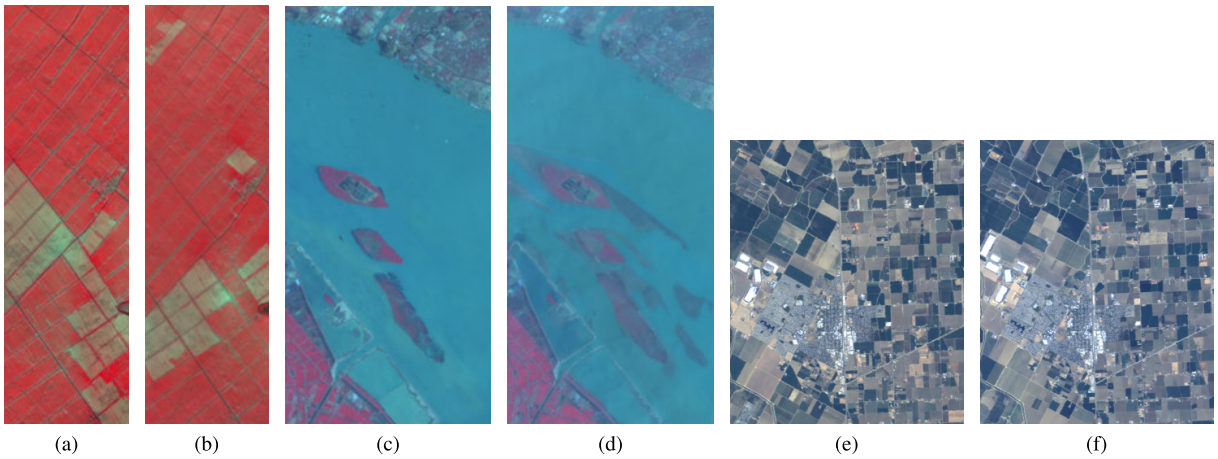


Fig. 7. Pseudocolor images of the datasets. (a) Image T_1 of Farmland. (b) Image T_2 of Farmland. (c) Image T_1 of River. (d) Image T_2 of River. (e) Image T_1 of Bay Area. (f) Image T_2 of Bay Area.

learn the intrinsic correlation between the features, yielding more discriminative change representations.

D. Prediction and Loss Function

For the training batch with the size of B, the CLS $\in \mathbb{R}^{B\times d}$ token from the final transformer encoder layer is fed into the classification head to obtain the detection results. The classifier is comprised of a linear layer and softmax function

$$Y_{\text{pred}} = \text{softmax}(\text{CLS} \cdot W_o) \tag{20}$$

where $Y_{\text{pred}} \in \mathbb{R}^{B\times 2}$ denotes the predictions for the current batch of samples, and W_o is the learned weight of the linear layer. The binary cross-entropy (BCE) loss function is widely used to solve binary classification problems. The CD task involves determining whether each pixel in one image is the same as the corresponding pixel in the other, which can be achieved using binary classification algorithms. Consequently, the BCE loss is adopted to optimize our network. The loss function can be expressed as follows:

$$L_{\text{BCE}} = -\sum_{i=1}^{B}\left(Y_{\text{gt}}^{i}\log Y_{\text{pred}}^{i} + \left(1-Y_{\text{gt}}^{i}\right)\log\left(1-Y_{\text{pred}}^{i}\right)\right) \tag{21}$$

where Y_{gt}^{i} and Y_{pred}^{i} denote the ground truth and prediction of ith given sample of the current training batch. Moreover, the labels for the unchanged and changed samples are set to 0 and 1, respectively.

III. Experiment Results and Analysis

A. Datasets

We utilize three datasets to validate the efficacy of our proposed approach. The first two datasets are the Farmland[1] and River[2] datasets, acquired by the Hyperion sensor aboard Earth Observing-1 (EO-1). The Hyperion sensor is designed to operate within the spectral wavelength range of 0.4–2.5 μm with a spectral resolution of 10 nm. Additionally, the sensor provides a spatial resolution of 30 m. Although the datasets are captured in 242 spectral bands, some bands are affected by noise during the collection, acquisition, and transmission processes. Therefore, the polluted bands have been removed in advance from the two public datasets. The last Bay Area[3] dataset is captured by the Airborne Visible InfraRed Imaging Spectrometer (AVIRIS). The AVIRIS sensor collects spectral information in the wavelength range of 0.4–2.5 μm and produces 224 continuous spectral bands. The spectral and spatial resolution are 10 and 20 m, respectively.

1) Farmland: As shown in Fig. 7(a) and (b), the first dataset covers an area of farmland in the city of Yancheng, Jiangsu Province, China. Two HSIs were acquired on May 3, 2006, and April 23, 2007. The size of each image is 420 × 140, and 154 spectral bands are retained for CD. This dataset contains 40 417 unchanged pixels and 18 383 changed pixels.

2) River: As shown in Fig. 7(c) and (d), the second dataset was collected on May 3, 2013, and December 31, 2013, respectively. The size of the images is 463 × 241 with 198 spectral bands. The main change is the substance of a river in Jiangsu Province, China. In this dataset, there are 101 885 unchanged pixels and 9698 changed pixels.

3) Bay Area: As shown in Fig. 7(e) and (f), the third dataset covers the city and farmland of California, USA. The images were taken in 2013 and 2015, which include 224 spectral bands with a spatial dimension of 600 × 500 pixels. There are 34 211 unchanged pixels and 39 270 changed pixels, and the rest of the dataset is unlabeled.

B. Evaluation Measures

In this study, we employ overall accuracy (OA), the Kappa coefficient, Sensitivity, Specificity, and F1 score as quantitative measures to evaluate the performance of our proposed method. Kappa is used to measure the consistency of the model's predicted results with the results of real changes. Sensitivity and Specificity measure the model's ability to correctly detect positive changes and identify unchanged areas, respectively. F1 score provides a balanced evaluation of a model's performance

[1]Available at rslab.ut.ac.ir.
[2]Available at http://crabwq.github.io.
[3]Available at https://citius.usc.es/investigacion/datasets/hyperspectral-change-detectiondataset.

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by considering both precision and recall

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \tag{22}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-p_e}{1-p_e} \tag{23}$$

$$p_e = \frac{(\mathrm{TP}\times\mathrm{TN})+(\mathrm{TP}\times\mathrm{TN})+(\mathrm{TP}\times\mathrm{TN})+(\mathrm{TP}\times\mathrm{TN})}{(\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN})^2} \tag{24}$$

$$\mathrm{Sensitivity} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{25}$$

$$\mathrm{Specificity} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}} \tag{26}$$

$$F1 = \frac{2\times\mathrm{TP}}{2\times\mathrm{TP}+\mathrm{FN}+\mathrm{FP}}. \tag{27}$$

C. Compared Methods and Experimental Setup

The following methods are selected to verify the effectiveness of our proposed MFCEN method.

1) The recurrent convolutional neural network (ReCNN) [20] involves bringing together a dilated CNN and an RNN into one end-to-end framework.
2) Re3FCN [19] integrates 3-D FCN and ConvLSTM to exact spatial-spectral features and change information, respectively.
3) CSANet [23] introduces cross-temporal interaction symmetric attention to enhance feature representation and discrimination ability in the Siamese-based 2-D CNN structure.
4) CDFormer [31] utilizes transformer encoders to exact the long-range spatial-temporal dependencies from the concatenated pixel sequences of each bitemporal image.
5) SST-Former [30] utilizes the transformer architecture to extract spatial-spectral features and temporal dependency subsequently.
6) ML-EDAN [39] employs the LSTM technique to capture temporal dependencies within hierarchical features, which are obtained by through dual encoder–decoder architecture.

D. Experimental Results

The MFCEN model is developed and executed on the Pytorch framework, utilizing an NVIDIA GeForce 1080Ti GPU for training. The initial learning rate and batch size are set to 1e-4 and 64, respectively. To optimize the model's performance, the learning rate is decayed every ten epochs based on the Step Learning Rate (StepLR) scheduler method, and Adam is used as an optimizer. Each branch of the model contains four layers of ResBlock. It should be noted that several parameters are adjusted to achieve optimal results on the three datasets, and these variations have been summarized in Table I for reference. In the experiments, we randomly stratify and select 5% of the data as the training samples. The training process is repeated six times, and the average of the results is considered as the final outcome.

TABLE I
EXPERIMENTAL PARAMETERS FOR THREE DATASETS

	Patch size	Layer	Head	Training epochs
Farmland	5	3	1	150
River	3	2	1	50
Bay Area	5	2	1	100

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON FARMLAND DATASETS

	OA	Kappa	Sensitivity	Specificity	F1
ReCNN	0.9459	0.8745	0.9189	0.9581	0.9141
Re3FCN	0.9623	0.9122	0.9399	0.9724	0.9397
CSANet	0.9620	0.9118	0.9411	0.9702	0.9395
CDFormer	0.9670	0.9229	0.9394	**0.9796**	0.9468
SST-Former	0.9507	0.8856	0.9247	0.9626	0.9215
ML-EDAN	0.9622	0.9121	0.9416	0.9715	0.9396
Proposed	**0.9690**	**0.9278**	**0.9479**	0.9786	**0.9503**

1) Farmland: The visual and quantitative performance of different methods is shown in Fig. 8 and Table II, respectively. The edges of the changed regions are blurred in the change maps of ReCNN and SST-Former, and there are even massive red false detections in the upper part of ReCNN. Therefore, ReCNN achieves the worst results in all evaluation metrics, and SST-Former performs slightly better. In the change map of Re3FCN, fewer miss and false detections are presented at the edge of the changed areas, which is also obviously better than the previous two methods in all evaluation indicators. Compared to Re3FCN, CSANet exhibits a higher count of false detections in red than miss detections in green, but their quantitative performance is almost the same. Although there are also some green and red spots in the detection result of ML-EDAN, it achieves the second-best Sensitivity, showing its relatively strong ability to detect the change region. CDFormer and our proposed method exhibit comparable and superior visual performance. However, a closer inspection reveals that CDFormer displays a higher count of red false detections in the lower middle area of the image. Correspondingly, our proposed method achieves the highest OA, Kappa, Sensitivity, and F1 score among all the methods, measuring 96.90%, 0.9278, 0.9485, and 0.9497, respectively. Only Specificity is slightly less than CDFormer by 0.0010, thus achieving the second-best performance in identifying unchanged regions. The proposed method's superior performance over CDFormer is attributed to the simultaneous extraction of change features and semantic features of bitemporal images with interaction and enhancement at each level.

2) River: Fig. 9 visually demonstrates that ReCNN exhibits the poorest performance, with numerous miss detections in green. This is further supported by the lowest sensitivity values presented in Table III, which are only 0.6716 and are substantially behind the penultimate method. Re3FCN, CSANet, and SST-Former exhibit similar visual and OA performance. However, upon comparing the other metrics, the sensitivity and F1 score of CSANet are lower than the other two methods at 0.6925 and 0.7614, but its Specificity is

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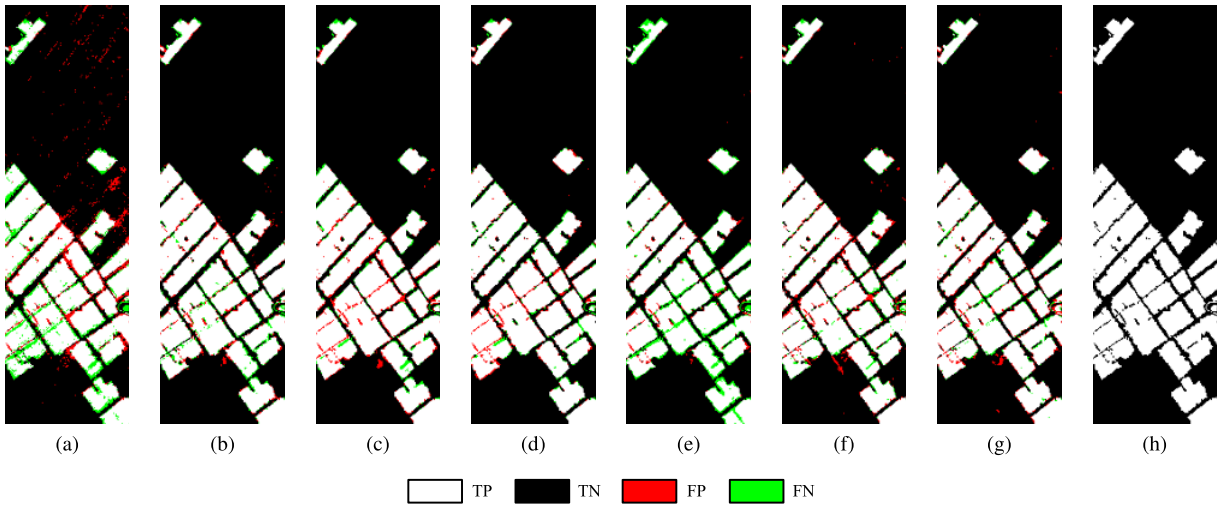


Fig. 8. Detection results for Farmland. (a) ReCNN. (b) Re3FCN. (c) CSANet. (d) CDFormer. (e) SST-Former. (f) ML-EDAN. (g) Proposed. (h) Ground Truth.

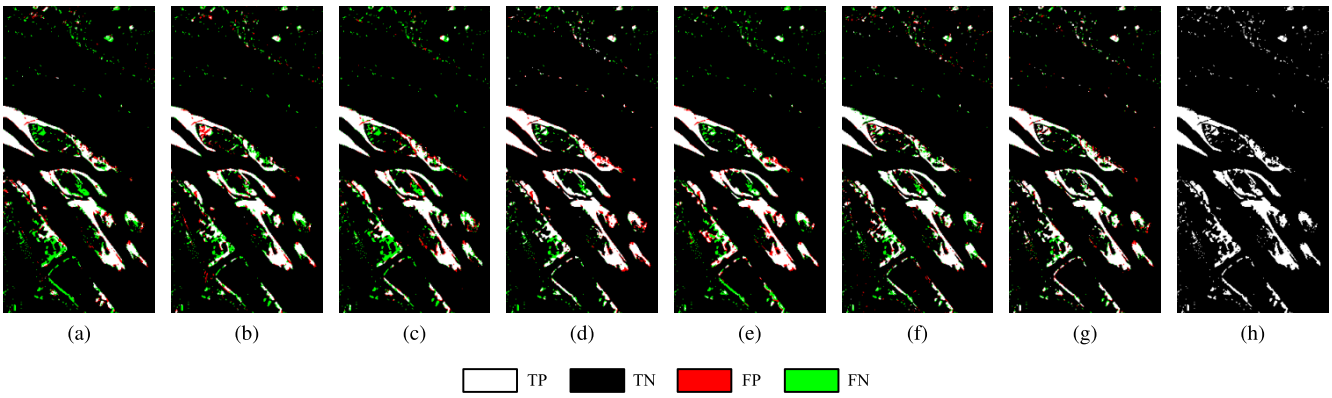


Fig. 9. Detection results for River. (a) ReCNN. (b) Re3FCN. (c) CSANet. (d) CDFormer. (e) SST-Former. (f) ML-EDAN. (g) Proposed. (h) Ground Truth.

TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON RIVER DATASETS

	OA	Kappa	Sensitivity	Specificity	F1
ReCNN	0.9603	0.7241	0.6716	0.9877	0.7453
Re3FCN	0.9626	0.7485	0.7148	0.9862	0.7687
CSANet	0.9623	0.7411	0.6925	0.9880	0.7614
CDFormer	0.9724	0.8217	0.8134	0.9876	0.8367
SST-Former	0.9632	0.7460	0.7139	0.9870	0.7656
ML-EDAN	0.9704	0.8037	0.7738	**0.9891**	0.8197
Proposed	**0.9763**	**0.8491**	**0.8526**	0.9881	**0.8621**

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON BAY AREA DATASETS

	OA	Kappa	Sensitivity	Specificity	F1
ReCNN	0.9724	0.9445	0.9680	0.9774	0.9740
Re3FCN	0.9907	0.9814	0.9894	0.9922	0.9913
CSANet	0.9911	0.9822	0.9900	0.9924	0.9917
CDFormer	0.9849	0.9697	0.9825	0.9877	0.9859
SST-Former	0.9732	0.9462	0.9642	0.9834	0.9746
ML-EDAN	0.9867	0.9733	0.9857	0.9879	0.9875
Proposed	**0.9926**	**0.9851**	**0.9919**	**0.9934**	**0.9931**

again slightly higher. ML-EDAN has fewer false detections in red than the previous methods, and its quantitative metrics are significantly better. Its specificity performs best among all methods, indicating that ML-EDAN can better identify unchanged regions. CDFormer has achieved the second-best performance among all comparison methods with an OA of 97.24%, a Kappa of 0.8217, a Sensitivity of 0.8134, and an F1 score of 0.8367. This highlights the effectiveness of temporal dependency during feature extraction in enhancing the accuracy of CD. Moreover, our proposed method achieves the best detection results, with the fewest green miss detections in the lower left part of our change map than other methods. Our detection accuracy outperforms CDFormer by 0.39%, 0.0274, 0.0392, 0.0005, and 0.0254 on OA, Kappa, Sensitivity, Specificity, and F1, respectively. This improvement can be attributed to our proposed method's ability better to learn change rules from multilevel enhanced change features.

3) Bay Area: Fig. 10 and Table IV illustrate the detection results of various methods on the Bay Area dataset. ReCNN and SST-Former achieve the worst performance in both visual and quantitative metrics, which lag significantly behind the other methods in all metrics. In the change map of Re3FCN, some green miss detections are still presented in the right areas. However, there are almost no red false detections, so all its evaluation metrics are improved with the previously mentioned methods. For the detection results of CSANet, only a few miss and false detections in the image, so it

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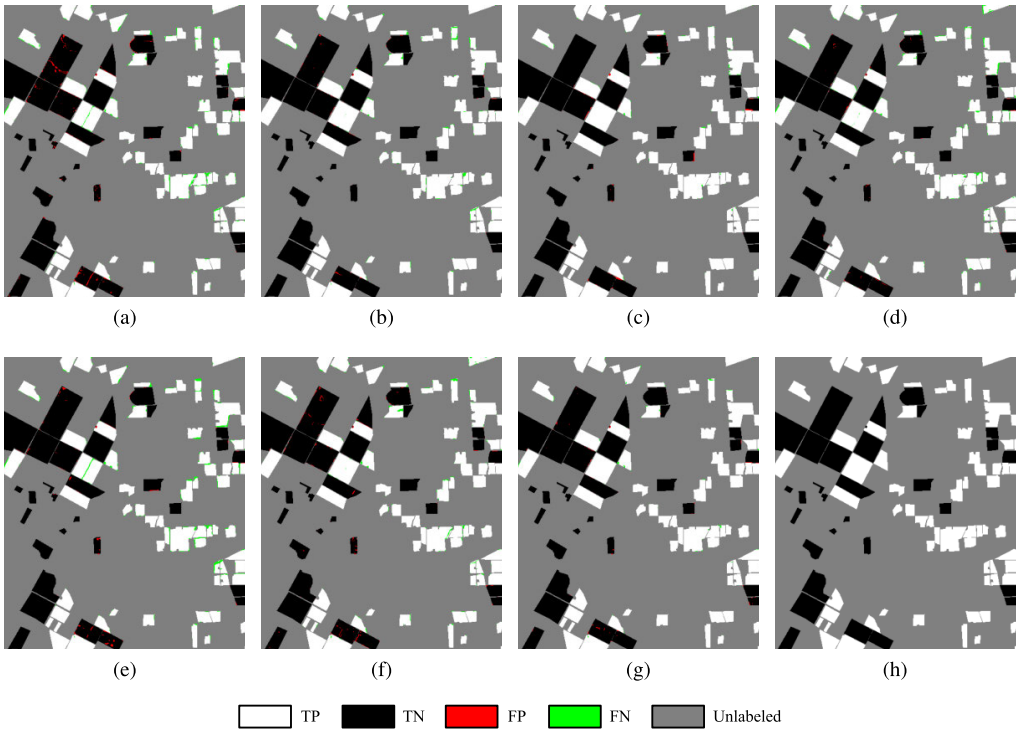


Fig. 10. Detection results for Bay Area. (a) ReCNN. (b) Re3FCN. (c) CSANet. (d) CDFormer. (e) SST-Former. (f) ML-EDAN. (g) Proposed. (h) Ground Truth.

achieves the second-best performance among all the methods. CDFormer and ML-EDAN produce almost the same area of false detection or miss detection. Nevertheless, ML-EDAN can detect better the small and irregular change regions in the upper right corner of the image, which makes ML-EDAN have better quantitative performance, 0.18%, 0.0036, 0.0032, and 0.0016 higher in OA, Kappa, Sensitivity, and F1-score. Our proposed MFCEN method demonstrates excellent detection capability in certain irregular small change regions, and there are almost no green miss detections or red false detections. Furthermore, MFCEN attains the highest detection results with OA, Kappa, Sensitivity, Specificity, and F1-score being 99.26%, 0.9851, 0.9919, 0.9934, and 0.9931, indicating an excellent ability to extract more distinctive features for CD. Our proposed MFCEN leverages multilevel features, enabling the acquisition of low-level spatial details and high-level semantic information, resulting in improved detection performance in small, irregular regions.

E. Ablation Study

1) TCIB: In our proposed method, we design a TCIB to directly capture and enhance the change features. To verify the effectiveness of TCIB, comparative experiments are performed on three datasets. From Table V, we can observe that the inclusion of TCIB in Model (2) results in remarkable performance, particularly on the Hermiston and River datasets, compared to Model (1) without TCIB. Furthermore, the performance of Model (4) surpasses Model (3) after the addition of TCIB. This indicates that our proposed TCIB can enhance the integration of temporal change information within different branches, improving CD performance.

2) MFFM: The proposed method aggregates multilevel enhanced change features by adopting the MFFM based on the transformer encoder. To evaluate the effect of the MFFM, we trained the proposed MFCEN method under different conditions. Notably, we utilized multiple fully connected layers in Model (1) and Model (2) to process features of different levels, as described in [38]. Table V illustrates that the integration of multilevel features by the MFFM leads to a significant improvement in performance for Model (3) compared to Model (1). Moreover, the comparison between Model (2) and Model (4) suggests that MFFM facilitates the aggregation of multilevel change information, resulting in enhanced detection performance.

TABLE V
ABLATION STUDY OF EACH MODULE ON THREE DATASETS

	Model	(1)	(2)	(3)	(4)
	TCIB	×	✓	×	✓
	MFFM	×	×	✓	✓
Farmland	OA	0.9620	0.9677	0.9616	**0.9690**
	Kappa	0.9112	0.9250	0.9107	**0.9278**
Hermiston	OA	0.9031	0.9661	0.9580	**0.9678**
	Kappa	0.7510	0.9023	0.8796	**0.9070**
River	OA	0.9277	0.9729	0.9569	**0.9763**
	Kappa	0.6630	0.8337	0.7608	**0.8491**

3) TAEM: In the proposed method, we introduce a temporal attention enhanced module to enhance the change information in bitemporal semantics and change features. To verify the importance of TAEM, we evaluate MFCEN with and without TAEM, denoted as w-TAEM and w/o-TAEM. Notably, the w/o-TAEM structure uses simple mathematical operations such as subtraction and addition to achieve the interaction between the change and bitemporal features. The experiment

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TABLE VI
ABLATION STUDY OF TEMPORAL ENHANCED ATTENTION MODULE ON THREE DATASETS

Method	China		Hermiston		River	
	OA	Kappa	OA	Kappa	OA	Kappa
w/o-TAEM	0.9664	0.9219	0.9661	0.9025	0.9681	0.8087
w-TAEM	**0.9690**	**0.9278**	**0.9678**	**0.9070**	**0.9763**	**0.8491**

TABLE VII
RESULTS OF DIFFERENT METHODS TO GET FUSED DIFFERENCE IMAGE ON THREE DATASETS

	Method	w-Add	w-Sub	w-Cat	w-BIFS
Farmland	OA	0.9624	0.9634	0.9634	**0.9690**
	Kappa	0.9128	0.9146	0.9153	**0.9278**
Hermiston	OA	0.9583	0.9622	0.9642	**0.9678**
	Kappa	0.8808	0.8911	0.8972	**0.9070**
River	OA	0.9613	0.9745	0.9691	**0.9763**
	Kappa	0.7680	0.8368	0.8118	**0.8491**

results are shown in Table VI. It can be seen that w-TAEM achieves superior performance across all datasets, especially in the River dataset. This indicates that our proposed TAEM effectively establishes the relationship between bitemporal semantics and change features, improving the performance of CD.

F. Discussion

1) BIFS: Traditionally, the difference image is obtained by addition (Add), subtraction (Sub), or concatenation (Cat) in HSI CD. To evaluate the effectiveness of BIFS, we conduct experiments by replacing BIFS with each of the methods mentioned above. Table VII shows the detection results of different operations on three datasets. It can be seen that w-Add yields the worst results because it struggles to capture the differences between bitemporal images. While w-Sub and w-Cat perform similarly on the Farmland dataset, w-Sub is slightly worst than w-Cat on the Hermiston dataset and significantly outperforms w-Cat on the River dataset. This shows the lack of robustness of these methods on different datasets. Cat can retain more not only original information but also more noise; Sub can directly obtain difference information but may struggle to preserve the characteristics of each bitemporal image. Our proposed BIFS leverages band-by-band convolution operations to leverage the spatial information of each band, generating fusion bands that contain both difference and original image information. Therefore, our proposed BIFS achieves the best performance on all datasets.

2) CBAM-BP: In our proposed method, the CBAM with bilinear pooling (CBAM-BP) is introduced to fully mine the spectral information in HSIs. To further assess the efficacy of CBAM-BP, we compare the experimental results of MFCEN with CBAM-BP (w-CBAM-BP), MFCEN with squeeze and excitation [41] block (w-SE), MFCEN with CBAM (w-CBAM), and MFCEN without any attention mechanism (w/o-CBAM-BP). Table VIII demonstrates that the model w/o-CBAM-BP achieves the lowest performance,

TABLE VIII
RESULTS OF DIFFERENT ATTENTIONS ON THREE DATASETS

	Method	w/o-CBAM-BP	w-SE	w-CBAM	w-CBAM-BP
Farmland	OA	0.9674	0.9685	0.9682	**0.9690**
	Kappa	0.9246	0.9267	0.9259	**0.9278**
Hermiston	OA	0.9648	0.9661	0.9653	**0.9678**
	Kappa	0.8996	0.9026	0.9001	**0.9070**
River	OA	0.9739	0.9751	0.9748	**0.9763**
	Kappa	0.8394	0.8436	0.8439	**0.8491**

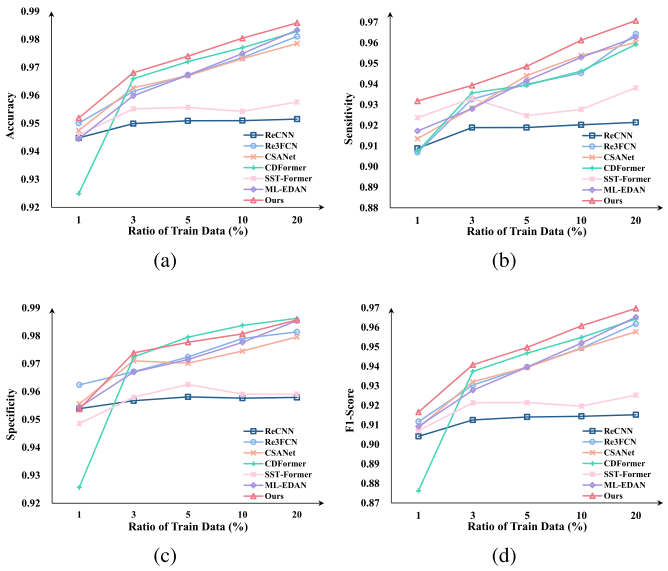


Fig. 11. Quantitative results of different methods with various training size on Farmland dataset. (a) OA. (b) Sensitivity. (c) Specificity. (d) F1-Score.

highlighting the crucial role of the attention mechanism in enhancing feature discriminability and improving detection accuracy. The models w-CBAM and w-SE modules exhibit similar performance, with w-SE performing slightly better due to its emphasis on channel information, which is advantageous for extracting spectral information from hyperspectral data. Our proposed CBAM-BP, which utilizes bilinear pooling to enhance CA for the rich spectral information of hyperspectral data, achieves the optimal performance among all techniques.

3) Ratio of Training Samples: To evaluate the robustness and generalization ability of the proposed method, we perform experiments with varying ratios of training data. The results of the Farmland dataset are illustrated in Fig. 11. As observed, our proposed MFCEN method obtains the best performance in OA, Sensitivity, and F1-Score across all ratios of training samples. While our method does not hold the lead in Specificity, it is only slightly worse than CDFormer at 5%, 10%, and 20% of the training data. Moreover, CDFormer performs very poorly when the training ratio is 1%. This arises from the stronger learning capability of our model by fully exploiting the multilevel enhanced change features. As the training ratio increases from 1% to 3%, the OA of all methods has been significantly improved, which shows that the detection ability of the current CD methods is lacking in small samples. However, with the increase of training data,

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TABLE IX
PARAMETERS AND AVERAGE COMPUTATION TIME OF DIFFERENT ALGORITHMS ON RIVER DATASET

Methods	Params(M)	Time(s)	
		Training	Inference
ReCNN	0.57	418.78	4.81
Re3FCN	0.97	137.96	4.00
CSANet	40.44	671.2	34.84
CDFormer	0.70	147.28	9.38
SST-Former	2.53	1098.30	41.03
ML-EDAN	112.69	2976.02	51.75
Proposed	154.73	434.98	45.28

the accuracy of ReCNN and SST-Former has improved overall but stabilized. This shows that ReCNN and SST-Former have a lower model performance ceiling. ML-EDAN performs the worst under 1% of training samples, but its metrics gradually surpass other comparison methods as train data increases, indicating ML-EDAN has overfitting problems. Re3FCN and CSANet exhibit a similar change trend, where the accuracy gradually rises as training samples increase. It should be noted that although the performance of CDFormer is better overall when the training data ratio is greater than 1%, its Sensitivity fluctuates wildly, indicating an unstable ability to identify change regions correctly.

4) Complexity and Time Cost: We compare the number of parameters and time cost of these methods on the River dataset. The experiments are conducted in Pytorch with Intel[4] Core[5] i9-9900K CPU at 3.60 GHz CPU and NVIDIA GeForce 1080Ti GPU. The experimental results are shown in Table IX. It can be seen that ReCNN, Re3FCN, and CDFormer exhibit the fewest parameters, but the time cost of ReCNN is significantly more than the other two methods. Although SST-Former has fewer parameters, its computation time is considerably high. CSANet has more than ten times the parameters of SST-Former but only requires slightly more than half of its training time. Among all methods, ML-EDAN has the second largest number of parameters but is the most time-consuming. Our proposed MFCEN has the most parameters but not the most time-consuming. Especially compared to ML-EDAN, MFCEN requires less than one-sixth of its training time despite having more parameters. Moreover, our proposed method achieves the best CD performance compared to all comparison methods.

In future studies, we will explore the implementation of pruning and parameter quantization techniques to reduce the number of parameters and time consumption while maintaining the accuracy of CD.

IV. CONCLUSION

In this article, we propose a novel multilevel features fused and change information enhanced neural network for HSI CD, where a TCIB is employed to directly exact and strengthen the change representations at each level, and the multilevel enhanced change features are fused by the transformer encoders. The low-level fine-grained details and high-level semantics are leveraged to generate the more discriminative change features, which contain rich semantic information and learn the intrinsic correlation between the features. By employing the designed cross-attention mechanism, this study effectively facilitates the interaction between bitemporal deep features and difference-image change representations, enabling more precise identification of the change area and the preservation of change regions' boundaries. The experiments on three public datasets validate that our proposed approach surpasses the current methods.

Although our proposed method has achieved considerable performance, it is notable that its superiority over other comparative methods on small training samples is not statistically significant. Therefore, in future work, we will persist in investigating techniques to enhance the efficacy of HSI CD methods when confronted with limited sample sizes.

[4]Registered trademark.
[5]Trademarked.

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Jigang Ding received the B.S. degree from the School of Electrical Engineering, Beijing Jiaotong University, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree in control theory and control engineering with Zhejiang University, Hangzhou, China.

His research interests include image processing and pattern recognition.



Xiaorun Li received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1992, and the M.S. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1995 and 2008, respectively.

Since 1995, he has been with Zhejiang University, where he is currently a Professor with the College of Electrical Engineering. His research interests include hyperspectral image processing, signal and image processing, and pattern recognition.



Shu Xiang received the B.S. degree from Zhejiang University, Hangzhou, China, in 2022, where he is currently pursuing the Ph.D. degree in control theory and control engineering.

His research interests include hyperspectral unmixing, image processing, and pattern recognition.



Shuhan Chen (Member, IEEE) received the B.S. degree from Ludong University, Yantai, China, in 2011, the M.S. degree from Liaoning Technical University, Huludao, China, in 2014, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2020.

She did research as a Visiting Faculty Assistant at the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, from 2018 to 2020. She did a Post-Doctoral position at the Department of Control Science and Engineering, Zhejiang University. She is currently an Assistant Professor with the Department of Control Science and Engineering, Zhejiang University. Her research interests include image registration, hyperspectral image processing, and pattern recognition.

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